WORKED SOLUTION

STEP II 2001, Question 6

Pure • Integration by Substitution

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/01-s2-q6

The key idea

The awkward-looking $\tan^{-1}\left(\frac{1-x}{1+x}\right)$ is really just $\frac{\pi}{4} - \tan^{-1} x$ in disguise: it is the tangent subtraction formula read backwards. Once that is spotted, part (i) becomes an integration by parts that lands exactly on the opening integral. Part (ii) looks unrelated, but the map $y = \frac{1-x}{1+x}$ swaps 0 and 1 and is its own inverse, so substituting it turns (ii) back into (i).

Solution

The opening integral

Dividing x^4 by $1 + x^2$ (or writing $x^4 = (x^4 - 1) + 1 = (x^2 - 1)(x^2 + 1) + 1$) gives

$$\frac{x^4}{1+x^2} = x^2 - 1 + \frac{1}{1+x^2}.$$

Hence

$$\int_0^1 \frac{x^4}{1+x^2}\,\mathrm{d}x = \left[\tfrac{1}{3}x^3 - x + \tan^{-1} x\right]_0^1 = \frac{1}{3} - 1 + \frac{\pi}{4} = \frac{\pi}{4} - \frac{2}{3},$$

as required.

Part (i)

Let $\theta = \tan^{-1} x$, so for $0 \le x \le 1$ we have $0 \le \theta \le \frac{\pi}{4}$. By the subtraction formula,

$$\tan\left(\frac{\pi}{4} - \theta\right) = \frac{\tan\frac{\pi}{4} - \tan\theta}{1 + \tan\frac{\pi}{4}\tan\theta} = \frac{1-x}{1+x}.$$

Ask yourself Does $\tan(\frac{\pi}{4} - \theta) = \frac{1-x}{1+x}$ really let us write $\tan^{-1}(\frac{1-x}{1+x}) = \frac{\pi}{4} - \theta$?

Only if $\frac{\pi}{4} - \theta$ lies in the range $(-\frac{\pi}{2}, \frac{\pi}{2})$ of $\tan^{-1}$. Here $0 \le \frac{\pi}{4} - \theta \le \frac{\pi}{4}$, so it does, and on the whole interval of integration

$$\tan^{-1}\Big(\frac{1-x}{1+x}\Big) = \frac{\pi}{4} - \tan^{-1} x.$$

Therefore

$$\int_0^1 x^3 \tan^{-1}\Big(\frac{1-x}{1+x}\Big)\,\mathrm{d}x = \frac{\pi}{4}\int_0^1 x^3\,\mathrm{d}x - \int_0^1 x^3 \tan^{-1} x\,\mathrm{d}x = \frac{\pi}{16} - J,$$

where $J = \int_0^1 x^3 \tan^{-1} x\,\mathrm{d}x$. Integrating by parts with $u = \tan^{-1} x$ and $\mathrm{d}v = x^3\,\mathrm{d}x$ (so $v = \frac{1}{4}x^4$ and $\mathrm{d}u = \frac{\mathrm{d}x}{1+x^2}$):

$$\begin{aligned} J &= \Big[\tfrac{1}{4}x^4 \tan^{-1} x\Big]_0^1 - \frac{1}{4}\int_0^1 \frac{x^4}{1+x^2}\,\mathrm{d}x \\ &= \frac{\pi}{16} - \frac{1}{4}\Big(\frac{\pi}{4} - \frac{2}{3}\Big) = \frac{1}{6}, \end{aligned}$$

using the opening result. Hence

$$\int_0^1 x^3 \tan^{-1}\Big(\frac{1-x}{1+x}\Big)\,\mathrm{d}x = \frac{\pi}{16} - \frac{1}{6}.$$

Part (ii)

? **Ask yourself** Where does the strange weight $\frac{(1-y)^3}{(1+y)^5}$ come from?

It is what $x^3\,\mathrm{d}x$ becomes under the same transformation. Substitute $y = \frac{1-x}{1+x}$, which solves to $x = \frac{1-y}{1+y}$: the map is its own inverse, and it sends $x = 0$ to $y = 1$ and $x = 1$ to $y = 0$. Then

$$1 - y = \frac{2x}{1+x}, \qquad 1 + y = \frac{2}{1+x}, \qquad \mathrm{d}y = -\frac{2}{(1+x)^2}\,\mathrm{d}x,$$

so

$$\frac{(1-y)^3}{(1+y)^5}\,\mathrm{d}y = \frac{8x^3}{(1+x)^3}\cdot\frac{(1+x)^5}{32}\cdot\Big(-\frac{2}{(1+x)^2}\Big)\mathrm{d}x = -\frac{x^3}{2}\,\mathrm{d}x,$$

and $\tan^{-1} y = \tan^{-1}(\frac{1-x}{1+x})$. The minus sign is absorbed by reversing the limits (y from 0 to 1 means x from 1 to 0):

$$\begin{aligned} \int_0^1 \frac{(1-y)^3}{(1+y)^5}\tan^{-1} y\,\mathrm{d}y &= \int_1^0 \Big(-\frac{x^3}{2}\Big)\tan^{-1}\Big(\frac{1-x}{1+x}\Big)\mathrm{d}x \\ &= \frac{1}{2}\int_0^1 x^3 \tan^{-1}\Big(\frac{1-x}{1+x}\Big)\mathrm{d}x = \frac{1}{2}\Big(\frac{\pi}{16} - \frac{1}{6}\Big) = \frac{\pi}{32} - \frac{1}{12} \end{aligned}$$

by part (i).

Checks

- Both answers are positive, as they must be (each integrand is non-negative on $[0,1]$): $\frac{\pi}{16} - \frac{1}{6} \approx 0.02968$ and $\frac{\pi}{32} - \frac{1}{12} \approx 0.01484$. Simpson's rule on the original integrands agrees to 12 decimal places.
- The answer to (ii) is exactly half that of (i), which you can sanity-check numerically before doing any algebra.
- Put $x = 0$ in the identity: $\tan^{-1} 1 = \frac{\pi}{4}$. Put $x = 1$: $\tan^{-1} 0 = 0$. Both match $\frac{\pi}{4} - \tan^{-1} x$.

Common slips

- Forgetting dy in (ii), or mishandling the sign: the factor $-\frac{2}{(1+x)^2}$ and the reversed limits cancel each other's minus signs.
- Writing $\tan^{-1}\left(\frac{1-x}{1+x}\right) = \frac{\pi}{4} - \tan^{-1} x$ without checking the range; for $x < -1$ it would be false (it is off by π).

Beyond STEP

The substitution $y = \frac{1-x}{1+x}$ is a Möbius transformation, a map of the form $\frac{ax+b}{cx+d}$. These maps form a group under composition, and this one has order two. The same trick, combined with the symmetry, gives the famous result $\int_0^1 \frac{\ln(1+x)}{1+x^2}\,\mathrm{d}x = \frac{\pi}{8}\ln 2$, which is worth trying next. Möbius maps are central in complex analysis; Needham's *Visual Complex Analysis* gives a beautiful geometric account of them.

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